Creditworthiness Report

Part 4 - Predicting Default Risk Project

ABSTRACT

Throughout this report, I will demonstrate every taken step to came up with the final classification model and will write down how many of the new customers would qualify for a loan.

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Problem Solving with Advanced Analytics-Udacity/Bertelsmann2021

Predicting Default Risk – Creditworthiness Report

Complete each section. When you are ready, save your file as a PDF document and submit it here:
https://classroom.udacity.com/nanodegrees/nd008/parts/11a7bf4c-2b69-47f3-9aec-108ce847f855/project

Step 1: Business and Data Understanding

Provide an explanation of the key decisions that need to be made. (250 word limit)

Key Decisions:

Answer these questions

1. What decisions needs to be made?
2. What data is needed to inform those decisions?

Due to a financial scandal that hit a competitive bank last week, our small bank suddenly has an **influx of new people applying for loans** at our bank instead of the other bank in our city. All of a sudden, our bank has nearly **500 loan applications** to process this week!

My manager sees this new influx as a great opportunity and wants me to figure out **how to process all of these loan applications within one week**. I should provide **a list of creditworthy customers** to my manager in the **next two days**!

Key Decisions:

1. **Decision**: From the **500** loan applications, determine **who is considered a creditworthy** customer for a loan and who is not?
2. **Needed Info**: The **probability percentage** for each customer to **be a creditworthy** customer or non-creditworthy customer.
3. **Analysis Type**: **Predictive** analysis to **predict the probability of being a creditworthy customer** for each customer has applied for a loan. That requires applying the "**Binary Classification Model**".
4. **Needed Data:** Any variables (**critical factors**) may affect **previous decision of credit approvals** from your past loan applicants the bank has ever completed "**Credit Application Result**". Below attributes are suspected:

- Account-Balance.
- Duration-of-Credit-Month.
- Payment-Status-of-Previous-Credit.
- Purpose.
- Credit-Amount.
- Value-Savings-Stocks.
- Length-of-Current-Employment.
- Installment-per-Cent.
- Guarantors.
- Duration-in-Current-Address.
- Most-Valuable-Available-Asset.
- Age-Years.
- Type-of-Apartment.
- Concurrent-Credits.
- No-of-Credit-at-this-Bank.

<u>Step 2</u>: Building the Training Set

Build your training set given the data provided to you. The data has been cleaned up for you already so you shouldn't ***need to convert any data fields to the appropriate data types.***

<u>Here are some guidelines to help guide your data cleanup:</u>

- For numerical data fields, are there any fields that highly-correlate with each other? The correlation should be at least .70 to be considered "high".
- Are there any missing data for each of the data fields? Fields with a lot of missing data should be removed
- Are there only a few values in a subset of your data field? Does the data field look very uniform (there is only one value for the entire field?). This is called "low variability" and you should remove fields that have low variability. Refer to the "Tips" section to find examples of data fields with low-variability.
- Your clean data set should have 13 columns where the Average of **Age Years** should be 36 (rounded up)

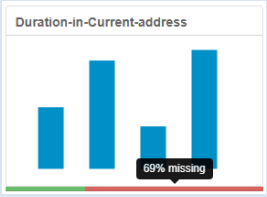
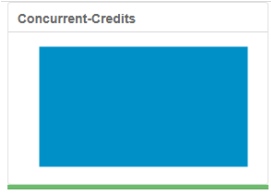
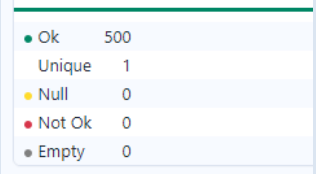
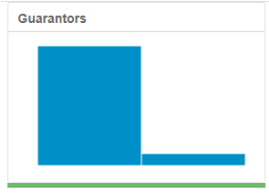
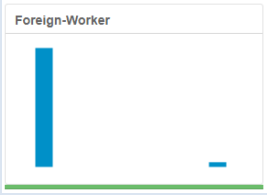
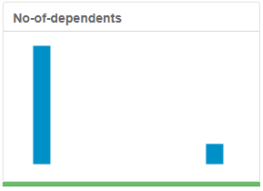
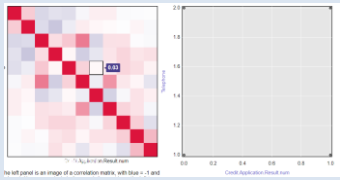
Note: *For the sake of consistency in the data cleanup process, impute data using the median of the entire data field instead of removing a few data points. (100 word limit)*

To achieve consistent results reviewers expect.

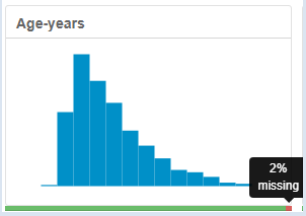
<u>Answer this question:</u>

- In your cleanup process, which fields did you remove or impute? Please justify why you removed or imputed these fields. Visualizations are encouraged.

- ***Removed Fields:***
  - I've decided to remove a total of **7 fields**. The decision back to **different reasons** as demonstrated below:

#	Field Name	Reason	Justification	Visual
1	**Duration-in-Current-address**	**Missing Data**	**High voulume** of data are **missing** (about **69%**)	
2	**Concurrent-Credits**	**Low Variability**	There is only **one value** for the entire field (Value = **Other Banks/Depts**)	
3	**Occupation**	**Low Variability**	There is only **one value** for the entire field (**Value = 1)**	
4	**Guarantors**	**Low Variability**	The data field heavily **skewed** towards "**None**"	
5	**Foreign-Worker**	**Low Variability**	The data field heavily **skewed** towards data values of "**1.0 to 1.1**"	
6	**No-of-dependents**	**Low Variability**	The data field heavily **skewed** towards data values of "**1.0 to 1.1**"	
7	**Telephone**	**No Correlation**	There **is no correlatio**n between "**Telephone**" and "**Acceptance Decision**". **Correlation = 0.030838** **P-Value = 0.7136766**	

- ***Imputed Field:***
  - I've decided to impute **1 field only** as demonstrated below:

#	Field Name	Reason	Justification	Visual
1	**Age-Year**	**Missing Data**	**Low voulume** of data are **missing** (about **2%**) Imputation done uisng "**Age Median = 33**" beacsue the data is "**right skewed**".	

Step 3: Train your Classification Models

First, create your Estimation and Validation samples where 70% of your dataset should go to Estimation and 30% of your entire dataset should be reserved for Validation. Set the Random Seed to 1.

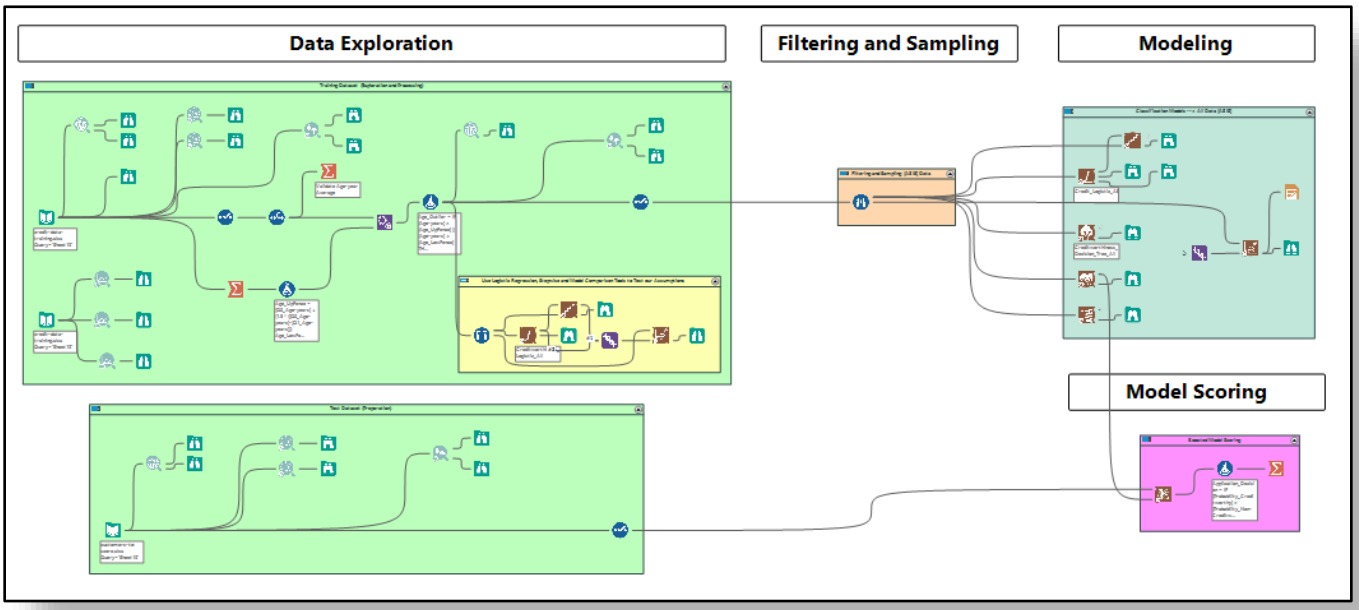
Create all of the following models: ***Logistic Regression***, ***Decision Tree***, ***Forest Model***, ***Boosted Model***

Answer these questions for ***each model*** *you created:*

- Which predictor variables are significant or the most important? Please show the p-values or variable importance charts for all of your predictor variables.
- Validate your model against the Validation set. What was the overall percent accuracy? Show the confusion matrix. Are there any bias seen in the model's predictions?

You should have four sets of questions answered. (500 word limit)

- ***My Alteryx Workflow:***



1. Logistic Regression (Adjusted based on Stepwise Model Result)

Model Result

(Against Training Dataset)

Coefficients:

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-2.9621914	6.837e-01	-4.3326	1e-05 ***
Account.BalanceSome Balance	-1.6053228	3.067e-01	-5.2344	1.65e-07 ***
Payment.Status.of.Previous.CreditPaid Up	0.2360857	2.977e-01	0.7930	0.42775
Payment.Status.of.Previous.CreditSome Problems	1.2154514	5.151e-01	2.3595	0.0183 *
PurposeNew car	-1.6993164	6.142e-01	-2.7668	0.00566 **
PurposeOther	-0.3257637	8.179e-01	-0.3983	0.69042
PurposeUsed car	-0.7645820	4.004e-01	-1.9096	0.05618 .
Credit.Amount	0.0001704	5.733e-05	2.9716	0.00296 **
Length.of.current.employment4-7 yrs	0.3127022	4.587e-01	0.6817	0.49545
Length.of.current.employment< 1yr	0.8125785	3.874e-01	2.0973	0.03596 *
Instalment.per.cent	0.3016731	1.350e-01	2.2340	0.02549 *
Most.valuable.available.asset	0.2650267	1.425e-01	1.8599	0.06289 .

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial taken to be 1)

Null deviance: 413.16 on 349 degrees of freedom
Residual deviance: 328.55 on 338 degrees of freedom
McFadden R-Squared: 0.2048, Akaike Information Criterion 352.5

Most Important Predictor Variables

- Account.Balance – "Some Balance" Value [P-Value = **1.65e-07**]
- Payment.Status.of.Previous.Credit – "Some Problems" Value [P-Value = **0.0183**]
- Purpose – "New car" Value [P-Value = **0.00566**]
- Credit.Amount [P-Value = **0.00296**]
- Length.of.current.employment – "< 1yr" Value [P-Value = **0.03596**]
- Instalment.per.cent [P-Value = **0.02549**]

Despite the **great P-Value** [**Less than 5%**] for above listed predictor variables, this model can explain only about **20%** of data [**R-Squared: 0.2048**]

Model Validation Result

(Against Validation Dataset)

Model	Accuracy	F1	AUC	Accuracy_Creditworthy	Accuracy_Non-Creditworthy
Credit_Log_Stepwise_All	0.7600	0.8364	0.7306	0.8762	0.4889
Credit_D.Tree_All	0.6667	0.7685	0.6272	0.7905	0.3778
Credit_Forest_All	0.7933	0.8681	0.7368	0.9714	0.3778
Credit_Boosted_All	0.7867	0.8632	0.7515	0.9619	0.3778

Confusion matrix of Credit_Log_Stepwise_All

	Actual_Creditworthy	Actual_Non-Creditworthy
Predicted_Creditworthy	92	23
Predicted_Non-Creditworthy	13	22

Measure	Creditworthy	Non-Creditworthy
Precision = (True / Class Predicted True Total)	92/115 = **80.0%**	22/35 = **63.0%**
Recall = (True / Class Actual Total)	92/105 = **87.6%**	22/45= **49.0%**
Error Rate = (Rev. False / Class Actual Total)	13/105 = **12.4%**	23/45 = **51.0%**
Overall Accuracy = (Predicted True Total / n)	(92 + 22) / 150 = **76.0%**	
Null Accuracy = (Dominated Class Actual Total / n)	(92 + 13) / 150 = **70.0%**	

- **Precision**: **80.0% TP Precision Vs. 63.0% TN Precision**
  1. **Low** precision %.
  2. **High** bias in prediction (**17.0**% difference).
- **Recall** Accuracy: **87.6% TPR Vs. 49.0% TNR**
  1. **Low** recall %.
  2. **High** bias in completeness (**38.6**% difference).
- **Error Rate**: **12.4% FNR Vs. 51% FPR**
  1. **High** error rate (especially for Negative Value)
- **Low Overall** Accuracy: as a value (**76.0%**) and in comparison with **Null acc.** (**70.0%**).

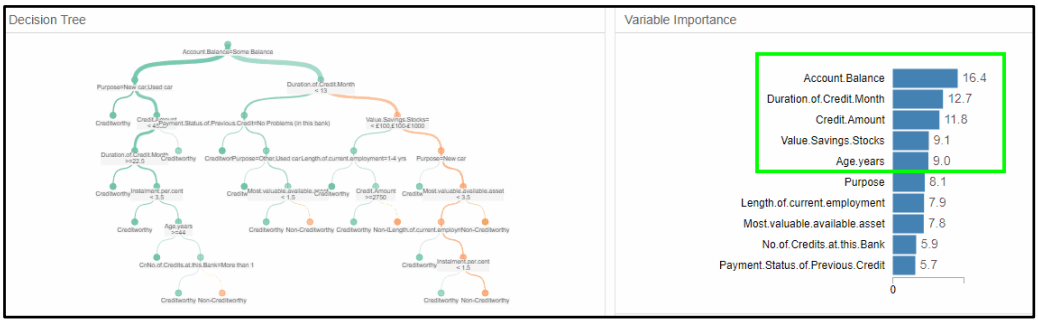
Decision

Not Accepted. Due to **1.** Low **Prediction**%. **2.** Low **Recall**%. **3.** High Prediction & Recall **Bias** and **4.** Low Overall **Accuracy**.

2. Decision Tree Model

Model Result

(Against Training Dataset)



Top 5 Important Predictor Variables [Weight about **59%** from all predictors]

- Account.Balance **[16.4 %]**
- Duration.of.Credit.Month **[12.7 %]**
- Credit.Amount **[11.8 %]**
- Value.Savings.Stocks **[9.1 %]**
- Age.Years **[9.0 %]**

Model Validation Result

(Against Validation Dataset)

Model	Accuracy	F1	AUC	Accuracy_Creditworthy	Accuracy_Non-Creditworthy
Credit_Log_Stepwise_All	0.7600	0.8364	0.7306	0.8762	0.4889
Credit_D.Tree_All	0.6667	0.7685	0.6272	0.7905	0.3778
Credit_Forest_All	0.7933	0.8681	0.7368	0.9714	0.3778
Credit_Boosted_All	0.7867	0.8632	0.7515	0.9619	0.3778

Confusion matrix of Credit_D.Tree_All

	Actual_Creditworthy	Actual_Non-Creditworthy
Predicted_Creditworthy	83	28
Predicted_Non-Creditworthy	22	17

Measure	Creditworthy	Non-Creditworthy
Precision = (True / Class Predicted True Total)	83/111 = **75.0%**	17/39 = **44.0%**
Recall = (True / Class Actual Total)	83/105 = **79.0%**	17/45 = **38.0%**
Error Rate = (Rev. False / Class Actual Total)	22/105 = **21.0%**	28/45 = **62.2%**
Overall Accuracy = (Predicted True Total / n)	(83 + 17) / 150 = **67.0%**	
Null Accuracy = (Dominated Class Actual Total / n)	(83 + 22) / 150 = **70.0%**	

- **Precision**: **75.0% TP Precision Vs. 44.0% TN Precision**
  1. **Low** precision %.
  2. **High** bias in prediction (**31.0**% difference).
- **Recall** Accuracy: **79.0% TPR Vs. 38.0% TNR**
  1. **Low** recall %.
  2. **High** bias in completeness (**41.0**% difference).
- **Error Rate**: **21.0% FNR Vs. 62.0% FPR**
  1. **High** error rate (especially for Positive Value)
- **Low Overall** Accuracy: as a value (**67.0%**) and in comparison with **Null acc.** (**70.0%**).

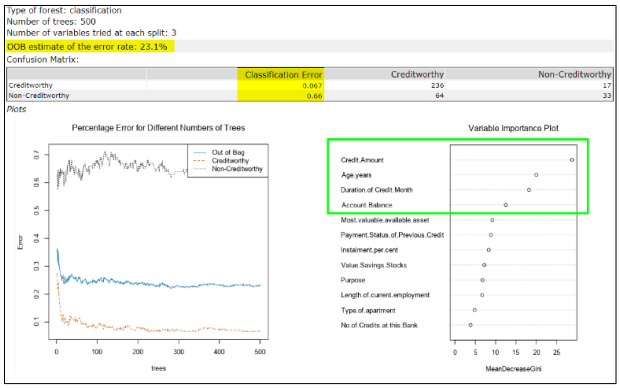
Decision

Not Accepted. Due to **1.** Low **Prediction**%. **2.** Low **Recall**%. **3.** High Prediction & Recall **Bias** and **4.** Low Overall **Accuracy**.

3. Forest Model

Model Result

(Against Training Dataset)



Top 4 Important Predictor Variables

- Credit.Amount
- Age.Years
- Duration.of.Credit.Month
- Account.Balance

This model has **high** error rate at prediction [**Error Rate: 23.1%**], especially for **Negative** Value "**Non-Creditworthy**" **66.0%**.

Model Validation Result

(Against Validation Dataset)

Model	Accuracy	F1	AUC	Accuracy_Creditworthy	Accuracy_Non-Creditworthy
Credit_Log_Stepwise_All	0.7600	0.8364	0.7306	0.8762	0.4889
Credit_D.Tree_All	0.6667	0.7685	0.6272	0.7905	0.3778
Credit_Forest_All	0.7933	0.8681	0.7368	0.9714	0.3778
Credit_Boosted_All	0.7867	0.8632	0.7515	0.9619	0.3778

Confusion matrix of Credit_Forest_All

	Actual_Creditworthy	Actual_Non-Creditworthy
Predicted_Creditworthy	102	28
Predicted_Non-Creditworthy	3	17

Measure	Creditworthy	Non-Creditworthy
Precision = (True / Class Predicted True Total)	102/130 = **78.5%**	17/20 = **85.0%**
Recall = (True / Class Actual Total)	102/105 = **97.0%**	17/45 = **38.0%**
Error Rate = (Rev. False / Class Actual Total)	3/105 = **3.0%**	28/45 = **62.2%**
Overall Accuracy = (Predicted True Total / n)	(102 + 17) / 150 = **79.3%**	
Null Accuracy = (Dominated Class Actual Total / n)	(102 + 3) / 150 = **70.0%**	

- **Precision**: **78.5% TP Precision Vs. 85.0% TN Precision**
  1. **High** precision %.
  2. **No** bias in prediction (**6.5% difference**).
- **Recall** Accuracy: **97.0% TPR Vs. 38.0% TNR**
  1. **Reasonable** recall %.
  2. **High** bias in completeness (**59.0% difference**).
- **Error Rate**: **3.0% FNR Vs. 62.2% FPR**
  1. **High** error rate (especially for Negative Value)
- **Good Overall** Accuracy: as a value (**79.3%**) and in comparison with **Null acc.** (**70.0%**).

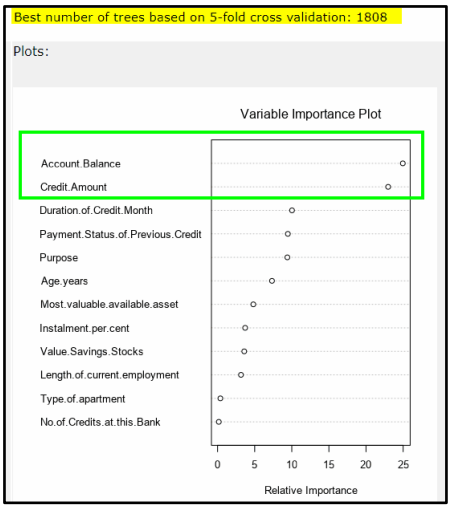
Decision

Accepted. Due to **1.** Good **Prediction**% and **Fairness 2.** Reasonable **Recall%**. **3.** Good Overall **Accuracy**. ***Should be compared to other "Accepted Models"***

4. Boosted Model

Model Result

(Against Training Dataset)



Top 2 Important Predictor Variables

- Account.Balance
- Credit.Amount

The model **re-run** based on **Best number of trees** based on 5-fold cross validation: **1808**

Model Validation Result

(Against Validation Dataset)

Model	Accuracy	F1	AUC	Accuracy_Creditworthy	Accuracy_Non-Creditworthy
Credit_Log_Stepwise_All	0.7600	0.8364	0.7306	0.8762	0.4889
Credit_D.Tree_All	0.6667	0.7685	0.6272	0.7905	0.3778
Credit_Forest_All	0.7933	0.8681	0.7368	0.9714	0.3778
Credit_Boosted_All	0.7867	0.8632	0.7515	0.9619	0.3778

Confusion matrix of Credit_Boosted_All

	Actual_Creditworthy	Actual_Non-Creditworthy
Predicted_Creditworthy	101	28
Predicted_Non-Creditworthy	4	17

Measure	Creditworthy	Non-Creditworthy
Precision = (True / Class Predicted True Total)	101/129 = **78.3%**	17/21 = **81.0%**
Recall = (True / Class Actual Total)	101/105 = **96.2%**	17/45 = **38.0%**
Error Rate = (Rev. False / Class Actual Total)	4/105 = **3.8%**	28/45 = **62.2%**
Overall Accuracy = (Predicted True Total / n)	(101 + 17) / 150 = **79.0%**	
Null Accuracy = (Dominated Class Actual Total / n)	(102 + 3) / 150 = **70.0%**	

- **Precision**: **78.3% TP Precision Vs. 81.0% TN Precision**
  1. **High** precision %.
  2. **No** bias in prediction (**2.7**% difference).
- **Recall** Accuracy: **96.2% TPR Vs. 38.0% TNR**
  1. **Reasonable** recall %.
  2. **High** bias in completeness (**58.2**% difference).
- **Error Rate**: **3.8% FNR Vs. 62.20% FPR**
  1. **High** error rate (especially for Negative Value)
- **Good Overall** Accuracy: as a value (**79.0%**) and in comparison with **Null acc.** (**70.0%**).

Decision

Accepted. Due to **1.** Good **Prediction**% and **Fairness 2.** Reasonable **Recall%**. **3.** Good Overall **Accuracy**. ***Should be compared to other "Accepted Models"***

Step 4: Writeup

Decide on the best model and score your new customers. For reviewing consistency, if Score_Creditworthy is greater than Score_NonCreditworthy, the person should be labeled as "Creditworthy"

Write a brief report on how you came up with your classification model and write down how many of the new customers would qualify for a loan. (250 word limit)

Answer these questions:

- Which model did you choose to use? Please justify your decision using **all** of the following techniques. Please only use these techniques to justify your decision:
    - Overall Accuracy against your Validation set
    - Accuracies within "Creditworthy" and "Non-Creditworthy" segments
    - ROC graph
    - Bias in the Confusion Matrices

Note: Remember that your boss only cares about prediction accuracy for Creditworthy and Non-Creditworthy segments.

- How many individuals are creditworthy?

Before you Submit

Please check your answers against the requirements of the project dictated by the rubric here. Reviewers will use this rubric to grade your project.

- ***Chosen Model:***
  - I've decided to choose the **"Forest Model"**.

- ***Decision Justification:***
  - Decision making criteria are set based on my boss request:
    1. **Best Overall Accuracy**.
    2. **Best Prediction accuracy "Precision%"** (for Creditworthy and Non-Creditworthy segments).
  - Thus, the **Forest Model** has been chosen because it **has the upper hand** over the **3 other models** in below criteria:

1. **Overall Accuracy against your Validation set**
   - Best Overall Accuracy (79.3%).

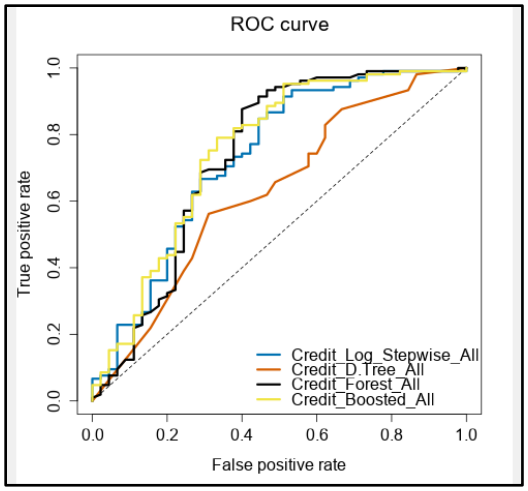
Model	Accuracy	F1	AUC	Accuracy_Creditworthy	Accuracy_Non-Creditworthy
Credit_Log_Stepwise_All	0.7600	0.8364	0.7306	0.8762	0.4889
Credit_D.Tree_All	0.6667	0.7685	0.6272	0.7905	0.3778
Credit_Forest_All	0.7933	0.8681	0.7368	0.9714	0.3778
Credit_Boosted_All	0.7867	0.8632	0.7515	0.9619	0.3778

2. **Accuracies within "Creditworthy" and "Non-Creditworthy" segments**
   - Putting "**Overall Accuracy**" into consideration, Forest Model has the Best Class Accuracies (97.1% "Creditworthy" and 38.0% "Non-Creditworthy").

Model	Accuracy	F1	AUC	Accuracy_Creditworthy	Accuracy_Non-Creditworthy
Credit_Log_Stepwise_All	0.7600	0.8364	0.7306	0.8762	0.4889
Credit_D.Tree_All	0.6667	0.7685	0.6272	0.7905	0.3778
Credit_Forest_All	0.7933	0.8681	0.7368	0.9714	0.3778
Credit_Boosted_All	0.7867	0.8632	0.7515	0.9619	0.3778

3. **ROC graph**
   - The biggest area under **ROC** Curve. That also appears clearly at **AUC**% (73.7%) corresponding to the best overall accuracy.

Model	Accuracy	F1	AUC	Accuracy_Creditworthy	Accuracy_Non-Creditworthy
Credit_Log_Stepwise_All	0.7600	0.8364	0.7306	0.8762	0.4889
Credit_D.Tree_All	0.6667	0.7685	0.6272	0.7905	0.3778
Credit_Forest_All	0.7933	0.8681	0.7368	0.9714	0.3778
Credit_Boosted_All	0.7867	0.8632	0.7515	0.9619	0.3778



4. **Bias in the Confusion Matrices**
   - As long as my boss only cares about prediction accuracy for Creditworthy and Non-Creditworthy segments, I'll evaluate the **Bias in Prediction Only**.
   - Although the "Forest Model" doesn't have the best Fairness across the other models, it has the **highest Overall Accuracy**.
   - Keeping high overall accuracy as major factor, the "Forest Model" has reasonable bias in prediction, especially when comparing with other models' accuracy.

Model	Overall Accuracy	Difference (Bias)	"Creditworthy" Precision	"Non-Creditworthy" Precision
Logistic Regression	**76.0%**	**17.0%**	**80.0%**	**63.0%**
Decision Tree	**67.0%**	**31.0%**	**75.0%**	**44.0%**
Forest Model	**79.3%**	**6.5%**	**78.5%**	**85.0%**
Boosted Model	**79.0%**	**2.7%**	**78.3%**	**81.0%**

Confusion matrix of Credit_Forest_All

	Actual_Creditworthy	Actual_Non-Creditworthy
Predicted_Creditworthy	102	28
Predicted_Non-Creditworthy	3	17

- ***Number of individuals that are creditworthy:*** ***408*** *persons are "**Creditworthy**".*

2 of 2 Fields ▾ | 2 records displayed

Record	Application_Decision	Count
1	Creditworthy	408
2	Non-Creditworthy	92

Thank You